INSIGHT MATRIX BD

Excel for Data Entry & Data Cleaning

A practical field guide to formulas, shortcuts and techniques for data analysts

Reliable reports start with tidy data. This guide shows you how to type data in consistently, recognise the usual problems, and turn a messy sheet into something you can safely analyse. It is written for students, aspiring data analysts and office professionals.

Your cleaning roadmap

Raw data ▶ Spot errors ▶ Clean ▶ Validate ▶ Analysis-ready

What you will practise

- Laying out tables so they filter and summarise without fuss.
- Using formulas and built-in tools to standardise untidy values.
- Investigating blanks and repeats instead of guessing at them.
- Validating your result while keeping the original data safe.

> **WORKING RULE**
> Never edit the raw data. Work on a copy or in helper columns, check what you produced, and write down every decision you made.

Inside this guide

#	Topic	What you get
01	Data entry best practices	Layout rules, Excel Tables, dropdown lists
02	Essential keyboard shortcuts	Fourteen time-savers with examples
03	Cleaning text with formulas	TRIM, PROPER, LEFT, MID, TEXTBEFORE and more
04	Dates, numbers & inconsistent values	Fixing text-numbers, dates and category variants
05	Missing data & duplicates	How to find, review and decide
06	Validation & error detection	Rules, checks and conditional formatting
07	Cleaning & preparation techniques	Splitting fields and choosing the right tool
08	Case study: online store orders	A full before-and-after walkthrough
09	Practice exercise & quick reference	Challenge, answer key and cheat sheets

01 Data Entry Best Practices

Most cleaning work exists because of how data was typed in the first place. A few habits at entry time save hours later.

Before and after: staff directory

Staff ID	Full Name	Joined	Department
S-001	tania sultana	2026-09-01	sales
S-002	Karim Uddin	05/09/26	Accounts
S-003	LIMA BEGUM		hr dept

▼ standardise names, dates and categories

Staff ID	Full Name	Joined	Department
S-001	Tania Sultana	2026-09-01	Sales
S-002	Karim Uddin	2026-09-05	Accounts
S-003	Lima Begum	(blank – review)	HR

Entry rules to live by

- **One row = one record, one column = one fact.** Do not pack two values into a single cell.
- **One header row only.** No merged cells and no empty rows in the middle of the data.
- **Right type for each field.** Dates as real dates, money as numbers, and IDs as text if leading zeros matter.
- **Make it an Excel Table.** Click inside the data, press `Ctrl+T`, and tick "My table has headers".
- **Use `Ctrl+E` (Flash Fill)** after typing one example of the pattern you want, then check every result before keeping it.

Build a Status dropdown

1. Select the Status cells.
2. Go to **Data → Data Validation** and set Allow to **List**.
3. In Source type `New,In Progress,Completed` and leave "In-cell dropdown" ticked.

> **DATA ANALYST TIP**
> Give every record a unique ID and, where practical, a validation rule to stop repeats. Excel Tables grow automatically as you add rows, so filters and formulas keep working.

> **CAUTION**
> A dropdown only controls what is typed from now on. Values that were already wrong, or that are pasted in later, stay wrong until you clean them.

02 Essential Keyboard Shortcuts

These are Windows desktop Excel shortcuts. Behaviour can vary with the active cell, your selection, or the mode Excel is in.

Shortcut	What it does	Quick example
Ctrl + T	Turns a range into an Excel Table	Format a selected dataset
Ctrl + Shift + L	Switches filters on or off	Add or remove filter arrows
Ctrl + E	Flash Fill from an example	Pull first names from full names
Ctrl + D	Fill down from the top cell	Copy a formula to the rows below
Ctrl + R	Fill right from the left cell	Copy a formula across columns
Ctrl + Enter	Enter one value in all selected cells	Type "Pending" into many cells at once
Alt + ↓	Opens a dropdown or filter list	Open a validation list
Ctrl + Arrow	Jumps to the edge of a data block	Reach the last row quickly
Ctrl + Shift + Arrow	Selects up to the edge of the block	Select a whole column of values
Ctrl + H	Find and Replace	Fix spelling variants of a status
Ctrl + Z	Undo	Reverse the last action
F4	Repeat action / cycle cell references	Switch A1 → $A$1 while editing
Ctrl + ;	Inserts today's date as a fixed value	Date-stamp an entry
Alt + =	AutoSum	Insert a SUM formula

Mini practice

- Type A, B, C in `A2:A4`, select the three cells and press `Ctrl+D`. Notice that everything becomes the top value, **A**.
- Select a small list, press `Ctrl+T`, then use the Status filter to show only "New".

> **CAUTION**
> `Ctrl+D` and `Ctrl+R` copy the top or left cell into your selection. They do not work out missing values for you, so always double-check the selected range first.

> **DATA ANALYST TIP**
> Learn three first: `Ctrl+T`, `Ctrl+Shift+L` and `Ctrl+Z`. Together they cover structure, searching and safety.

03 Cleaning Text with Formulas

Text is where most mess lives: stray spaces, odd capitals, wrong separators. A handful of functions handles the majority of it.

Example: tidy up supplier names

Raw name (A)	Formula (B)	Result
noor traders	`=PROPER(TRIM(A2))`	Noor Traders
DHAKA FRESH FOODS	`=PROPER(TRIM(A3))`	Dhaka Fresh Foods
aLi enterprise	`=PROPER(TRIM(A4))`	Ali Enterprise

Your text toolkit

Formula	What it does	Example result
`=TRIM(A2)`	Removes extra leading, trailing and repeated spaces	" hi there " → "hi there"
`=CLEAN(A2)`	Strips many non-printing characters	Depends on the source
`=SUBSTITUTE(A2,"-"," ")`	Swaps one piece of text for another	A-101 → A 101
`=UPPER(A2) / =LOWER(A2)`	Forces capitals or lowercase	bd → BD
`=LEFT(A2,3)`	First 3 characters	ORD-001 → ORD
`=RIGHT(A2,3)`	Last 3 characters	ORD-001 → 001
`=MID(A2,5,3)`	3 characters starting at position 5	ORD-001 → 001
`=LEN(A2)`	Counts characters, spaces included	"ABC " → 4
`=TEXTBEFORE(A2,"@")`	Text before a marker (newer Excel)	sara@mail.com → sara
`=TEXTAFTER(A2,"@")`	Text after a marker (newer Excel)	sara@mail.com → mail.com

Argument order: `MID(text, start, count)`, `LEFT/RIGHT(text, count)`, `SUBSTITUTE(text, old, new)`. TEXTBEFORE and TEXTAFTER need Microsoft 365 or a recent Excel release.

CAUTION

TRIM only removes ordinary spaces, not every non-breaking space. CLEAN removes many control characters, but not every invisible Unicode one. PROPER fixes capital letters, not spelling, and may not suit names like "McDonald".

DATA ANALYST TIP

Always clean into a new column. Compare it against the original, and delete nothing until your checks pass.

04 Dates, Numbers & Inconsistent Values

Numbers that are really text

A value like "1,200" may look numeric yet behave as text, so totals ignore it. Strip the comma, then convert.

Invoice	Raw amount	Formula	Result
INV-11	1,450	`=VALUE(SUBSTITUTE(B2,",",""))`	1450 (number)
INV-12	920	`=VALUE(SUBSTITUTE(B3,",",""))`	920 (number)
INV-13	N/A	`=IFERROR(VALUE(SUBSTITUTE(B4,",","")),"")`	Blank – flag it

A blank result hides why it is blank. Pair it with a status column such as:

```
=IFERROR(IF(VALUE(SUBSTITUTE(B2,",",""))>=0,"Valid number","Review"),"Invalid amount")
```

Common problems and their fixes

Problem	Technique	Remember
Date stored as text	`=DATEVALUE(A2)`	Only works if your regional settings recognise the format
Extra spaces in labels	`=TRIM(A2)`	Applies to category columns too
Currency symbols	`=VALUE(SUBSTITUTE(SUBSTITUTE(A2,"Tk",""),",",""))`	Match the symbol and decimal style of your source
Category spelling variants	`Ctrl+H or a mapping table`	Never merge labels that mean different things
Two facts in one cell	`Data → Text to Columns`	Choose the delimiter and read the preview first

DATE WARNING

`01/02/2026` could be 1 February or 2 January. Confirm the source format before converting. Also, changing a cell's number format does not reliably turn text into a true date or number.

DATA ANALYST TIP

Flag bad values (for example "Review amount") instead of quietly turning them into zero. Blank, zero, N/A and Unknown each mean something different.

05 Missing Data & Duplicate Records

Gaps and repeats are warning signs, not instructions to delete. Look first, then decide.

Sample to review: vendor list

Vendor ID	Name	Phone	Region
V-201	Anika Stores	01811111111	Dhaka
V-202	Babul Mart		Sylhet
V-202	Babul Mart		Sylhet
V-203	City Pharma	01833333333	
V-204	Delta Foods	01844444444	Khulna

How to investigate

- **Find blanks:** filter the column, or use Home → Find & Select → Go To Special → Blanks.
- **Flag repeated IDs:** `=COUNTIF($A$2:$A$100,A2)>1`. TRUE means the ID repeats, not that the whole row is a copy.
- **Remove Duplicates** only after you choose the right columns and confirm the repeats are not genuine separate transactions.
- **Fill down** only when the business logic says a blank means "same as above".
- **Never invent** phone numbers or categories. Use a "(missing)" label only if your reporting standard asks for one.

After review

Vendor ID	Name	Phone	Region	Decision
V-201	Anika Stores	01811111111	Dhaka	Keep
V-202	Babul Mart	(missing)	Sylhet	Repeated ID – investigate
V-202	Babul Mart	(missing)	Sylhet	Investigate, do not auto-delete
V-203	City Pharma	01833333333	(missing)	Confirm region
V-204	Delta Foods	01844444444	Khulna	Keep

> **DATA ANALYST TIP**
> COUNTIF spots repeated IDs only. To test a whole row, compare the relevant fields side by side or run Remove Duplicates with every column ticked.

06 Data Validation & Error Detection

Validation steers people towards correct entries, and checks catch the mistakes that still slip through.

Prevent and highlight errors

Goal	Set-up	Example
Dropdown	Data → Data Validation → List	New,In Progress,Completed
Whole number	Data Validation → Whole number	Quantity between 1 and 500
Decimal	Data Validation → Decimal	Discount between 0 and 1
Date range	Data Validation → Date	Order date inside the allowed period
Missing ID	=IF(A2="","Missing ID","OK")	Shows "Missing ID" when A2 is empty
Numeric check	=ISNUMBER(B2)	TRUE only for real numbers
Blank check	=ISBLANK(A2)	TRUE only for a truly empty cell

Highlight duplicate IDs with Conditional Formatting

1. Select `A2:A100`.
2. Choose **Home → Conditional Formatting → New Rule → Use a formula to determine which cells to format**.
3. Enter `=COUNTIF($A$2:$A$100,A2)>1`.
4. Pick a light red fill and click OK.

The dollar signs freeze the range `$A$2:$A$100`, while `A2` stays relative and shifts to A3, A4 and so on as Excel checks each row.

Colour key

RED = Review	Blank ID, repeated ID, invalid amount	**GREEN = Valid**	Required fields present; checks passed

> **CAUTION**
> Validation mainly works while people type. Pasted or imported data can slip past it, so run your checks on the finished dataset as well.

07 Cleaning & Preparation Techniques

Splitting full names into two columns

Full name	First	Last
Amina Rahman	Amina	Rahman
Rafiul Islam	Rafiul	Islam
Nusrat Jahan	Nusrat	Jahan

Option 1: Flash Fill. Type "Amina" in B2 and "Rahman" in C2, then press `Ctrl+E` in each column. Check every result, especially names with several parts.

Option 2: Text to Columns. Select the names, then Data → Text to Columns → Delimited → Space. Names with three or more parts will spill into extra columns, so read the preview.

Option 3: Formulas. `=TEXTBEFORE(A2," ")` and `=TEXTAFTER(A2," ")` take the text around the first space. They are simple and will not cope with complex multi-part names.

Which tool should I pick?

If you need...	Start with	Because...
A quick one-off pattern	Flash Fill	Fast and formula-free, but verify it
A split on a steady delimiter	Text to Columns	Built in, with a live preview
Cleaning that repeats every week	Power Query	Saved steps rerun on each new file
Rules you can audit in the cell	Formulas	Transparent and easy to trace

Handy reminders

- `Ctrl+H`: replace a known variant, but check the exact text first.
- Remove Duplicates: choose the right columns and keep a raw copy.
- Sort and Filter: isolate suspicious rows without deleting them.
- `Ctrl+T`: a structured table grows with your data and keeps filters working.

> **DATA ANALYST TIP**
> Power Query shines when the same import-and-clean routine is needed monthly or weekly. Build the steps once and refresh.

08 Case Study: Online Store Orders

A. Before: the raw export

Order ID	Customer	Date	Amount	Status
ORD-301	sara ahmed	2026-10-01	1,200	shipped
ORD-302	RAHIM KHAN	02/10/2026	850	Delivered
ORD-302	RAHIM KHAN	02/10/2026	850	Delivered
ORD-304	nUsRat jahan	2026-10-04	N/A	pending
ORD-305	Amina Rahman		500	In progress
ORD-306	Samiul Islam	2026/10/06	2,000	Shipped
ORD-307	Rafi Hasan	2026-10-08	900	in progress

B. The cleaning workflow

- **Protect the original.** Leave the raw sheet alone and work in a copy.
- **Customer:** `=PROPER(TRIM(B2))` gives "Sara Ahmed". Check any names that need special capitals.
- **Amount:** `=IFERROR(VALUE(SUBSTITUTE(D2,",","")),"")` returns 1200. "N/A" comes back blank and gets a "Review amount" flag.
- **Status:** use a mapping table or reviewed replacements: shipped → Shipped, pending → Pending, in progress → In Progress.
- **Dates:** confirm the source format, convert to real dates, then display as `yyyy-mm-dd`.
- **ORD-302 appears twice:** find out whether it is a double export or two real lines before removing anything. **ORD-305** has no date, so flag it and do not invent one.

C. After: reviewed working dataset

Order ID	Customer	Date	Amount	Status	Review
ORD-301	Sara Ahmed	2026-10-01	1200	Shipped	OK
ORD-302	Rahim Khan	2026-10-02*	850	Delivered	Duplicate ID
ORD-302	Rahim Khan	2026-10-02*	850	Delivered	Duplicate ID
ORD-304	Nusrat Jahan	2026-10-04	(blank)	Pending	Amount N/A
ORD-305	Amina Rahman	(blank)	500	In Progress	Missing date
ORD-306	Samiul Islam	2026-10-06	2000	Shipped	Verify date
ORD-307	Rafi Hasan	2026-10-08	900	In Progress	OK

** Use 2026-10-02 only if the source confirms DD/MM/YYYY. Otherwise leave it flagged.*

> **DATA ANALYST TIP**
> Final checklist: clear headers ✓ correct data types ✓ standard text ✓ missing values reviewed ✓ duplicates investigated ✓ invalid values flagged ✓ filters and validation tested ✓

09 Practice Exercise & Quick Reference

Your challenge: clean this mini dataset

ID	Name	Amount	Status
E-01	tania SULTANA	2,500	new
E-02	KARIM UDDIN	N/A	In Progress
E-02	KARIM UDDIN	1,100	in progress
E-04	Lima Begum	950	Completed
	Sohan Mia	700	new

Tasks: (1) clean names in a helper column, (2) convert valid amounts and flag N/A, (3) standardise status, (4) flag repeated IDs, (5) flag the missing ID. Keep the raw columns.

Answer key

- **Name:** `=PROPER(TRIM(B2))` → Tania Sultana, and so on down the column.
- **Amount:** `=IFERROR(VALUE(SUBSTITUTE(C2,",","")),"")` → 2500, blank, 1100, 950, 700. Add a separate validity check.
- **Repeated ID:** `=AND(A2<>"",COUNTIF($A$2:$A$6,A2)>1)` → TRUE for both E-02 rows, FALSE for the blank ID.
- **Missing ID:** `=IF(A2="","Missing ID","OK")` → flags the Sohan Mia row.
- **Status:** New, In Progress, In Progress, Completed, New.

Cheat sheets

Task	Formula
Trim spaces	`=TRIM(A2)`
Standard name case	`=PROPER(TRIM(A2))`
Convert numeric text	`=VALUE(A2)`
Flag repeated ID	`=COUNTIF($A$2:$A$100,A2)>1`
Check missing ID	`=IF(A2="","Missing ID","OK")`
Check numeric value	`=ISNUMBER(B2)`

Shortcuts: `Ctrl+T` Table • `Ctrl+Shift+L` Filters • `Ctrl+E` Flash Fill • `Ctrl+H` Replace • `Ctrl+Z` Undo • `Ctrl+;` Date • `Alt+=` AutoSum • `F4` Cycle references

> **CAUTION**
> Avoid these: overwriting raw data, treating blanks as zero, deleting repeated IDs unchecked, trusting ambiguous dates, and using IFERROR to hide a problem without a review flag.

Practise on copies of realistic data. Clean carefully, record your decisions, and every report you build will earn more trust.